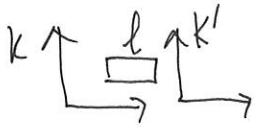
Little practice with Lorentz transformation

length contraction: which form to use

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} ct' \\ x' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}$$



The rod is stationary in K' ⇒ t' does not play role.

⇒ use $\left.\begin{aligned} x'_L &= \gamma(x_L - \beta ct) \\ x'_R &= \gamma(x_R - \beta ct) \end{aligned}\right\}$ both ends, x_L and x_R, are observed at the same time t.

$$\Rightarrow \Delta x' = x'_R - x'_L = \gamma(x_R - x_L) = \gamma \Delta x$$

$$\Rightarrow \underbrace{\Delta x}_{l} = \frac{\Delta x'}{\gamma} = \sqrt{1 - v^2/c^2}\, \underbrace{\Delta x'}_{l_0}$$

l_0 ← length in the rest frame K'

Time dilation: moving clock is always at $x' = 0$ in its rest frame K'.

⇒ we use $\left.\begin{aligned} ct_1 &= \gamma(ct'_1 + \beta x') \\ ct_2 &= \gamma(ct'_2 + \beta x') \end{aligned}\right\}$ $(ct_2 - ct_1) = \gamma(ct'_2 - ct'_1)$

$$\Rightarrow T = \gamma T' = \frac{1}{\sqrt{1 - v^2/c^2}} T'$$

Lorentz transformation as a rotation of Minkowski space-time

Look for the linear transformation (because uniform motion in K must remain uniform in K') that preserves the interval $s^2 = c^2t^2 - \vec{r}^{\,2}$.

$$ct' = mct + nx \qquad m, n, p, q \text{ are coefficients.}$$
$$x' = pct + qx$$

Require $c^2t^2 - x^2 = c^2t'^2 - x'^2 = (mct + nx)^2 - (pct + qx)^2$
$$= (m^2 - p^2)c^2t^2 - (q^2 - n^2)x^2 + 2(mn - pq)ctx$$

$\Rightarrow$ $m^2 - p^2 = 1$ $\Rightarrow$ let $m = ch\xi$, $p = -sh\xi$

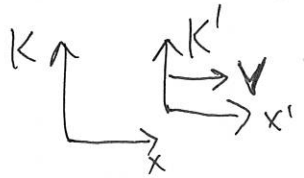
$q^2 - n^2 = 1$ $\Rightarrow$ $q = ch\psi$, $n = -sh\psi$

$mn - pq = 0 = -ch\xi\, sh\psi + sh\xi\, ch\psi = sh(\xi - \psi)$

$\Rightarrow \xi = \psi$

Then $\begin{cases} ct' = ch\psi\, ct - sh\psi\, x \\ x' = -sh\psi\, ct + ch\psi\, x \end{cases}$

ψ = rapidity.



Origin of K' ($x' = 0$) corresponds to $x = vt$ in K

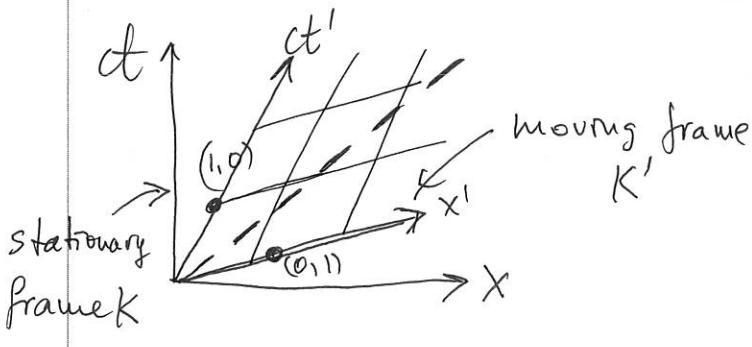
$\Rightarrow x' = 0 = -sh\psi\, ct + ch\psi\, vt$

$\Rightarrow th\psi = \frac{v}{c} = \beta$.

Thus $ch\psi = \frac{1}{\sqrt{1 - th^2\psi}} = \frac{1}{\sqrt{1-\beta^2}} = \gamma$

$sh\psi = th\psi\, ch\psi = \beta\gamma$

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} \; ; \; \begin{pmatrix} ct \\ x \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}$$

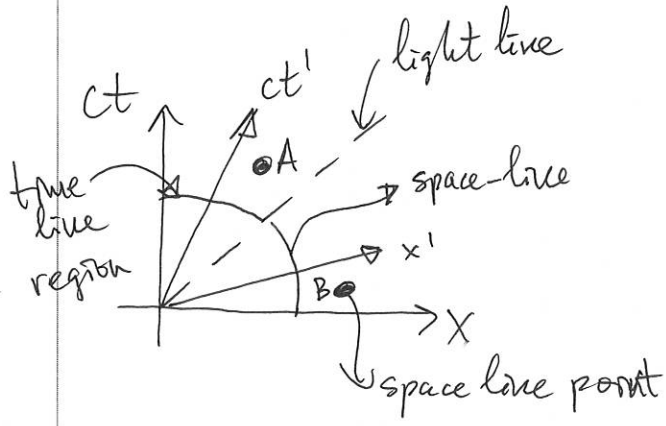


Consider $(ct', x') = (1,0)$

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma \\ \gamma\beta \end{pmatrix}$$

$$\gamma\beta = \sqrt{\frac{\beta^2}{1-\beta^2}}$$

$$(ct', x') = (0,1) \Rightarrow (ct, x) = \gamma(\beta, 1)$$



$ct > 0$ but $ct' < 0$

A: $ct > 0$ and $ct' > 0$

Velocity transformations

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$d(ct) = \gamma\, d(ct') + \gamma\beta\, dx'$$
$$dx = \gamma\beta\, d(ct') + \gamma\, dx'$$

$$dy = dy', \quad dz = dz'$$

$$v_x = \frac{dx}{dt} = c\,\frac{dx' + \beta c\, dt'}{\beta\, dx' + c\, dt'} = c\,\frac{\left(\frac{dx'}{dt'} + \beta c\right) dt'}{c\, dt'\left(1 + \frac{\beta}{c}\frac{dx'}{dt'}\right)} =$$

$$\left(\beta = \frac{v}{c}\right) \qquad = \frac{v_x' + v}{1 + \frac{v_x' v}{c^2}}$$

Recall Galilean result: $v_x = v_x' + v$

The difference become significant when $v_x', v \sim c$.

Suppose $v = \frac{3c}{4}$ and $v_x' = \frac{3c}{4}$

then $v_x\big|_{\text{Galilean}} = \frac{3c}{4} + \frac{3c}{4} = \frac{3}{2}c$

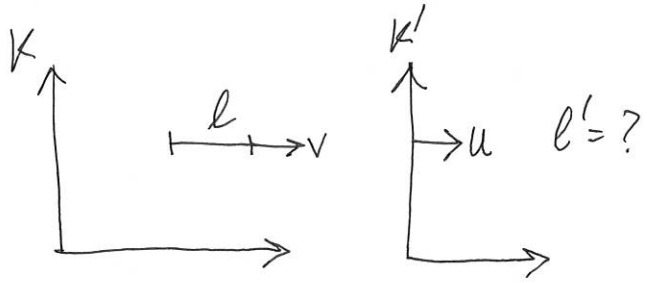
while $v_x\big|_{\text{Lorentz}} = \dfrac{\frac{3c}{4} + \frac{3c}{4}}{1 + \left(\frac{3c}{4c}\right)^2} = \dfrac{\frac{3}{2}c}{1 + \frac{9}{16}} = \frac{16}{25}\cdot\frac{3c}{2} =$

$$= \frac{24}{25}c.$$

$$v_y = \frac{dy}{dt} = \frac{c\, dy'}{\gamma(\beta\, dx' + c\, dt')} =$$

$$= \frac{c\, dy'}{\gamma c\, dt'\left(1 + \frac{\beta}{c}\frac{dx'}{dt'}\right)} = \frac{v_y'}{\gamma\left(1 + \frac{v_x' v}{c^2}\right)}$$

P: an object is measured to have length l in frame K where it moves with $\vec{v}$. What is its length l' in the frame K' which moves with $\vec{u}$ relative to K? Let $\vec{u} \parallel \vec{v}$.



Need to think of the rest frame K'' of the object!

$$l = \frac{l_0}{\gamma} = \sqrt{1 - v^2/c^2}\, l_0$$

$l'' = l_0$ = length in frame K'' where object is at rest.

K'' moves with $\vec{v}$ relative to K

Velocity of object on K is $v = v_x$.

$$v_x = \frac{v_x' + u}{1 + \frac{v_x' u}{c^2}}$$

where v_x' = velocity of object in K'

$$v_x' = \frac{v_x - u}{1 - \frac{v_x u}{c^2}} = \frac{v - u}{1 - \frac{vu}{c^2}}$$

$$\Rightarrow \text{in } K' \quad l' = \sqrt{1 - \frac{(v_x')^2}{c^2}}\, l_0$$

$$= \sqrt{1 - \left(\frac{v-u}{1 - \frac{vu}{c^2}}\right)^2 \frac{1}{c^2}}\; \frac{l}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$= \frac{\sqrt{\left(1 - \frac{vu}{c^2}\right)^2 - \left(\frac{v-u}{c}\right)^2}}{1 - \frac{vu}{c^2}}\; \frac{l}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$= \sqrt{1 - \cancel{\frac{2vu}{c^2}} + \frac{v^2u^2}{c^4} - \cancel{\frac{v^2 + u^2 - 2vu}{c^2}}}\; \frac{l}{\left(1 - \frac{vu}{c^2}\right)\sqrt{1 - \frac{v^2}{c^2}}}$$

Check:

(1) $u = 0 \Rightarrow l' = l$ since then $K' = K$.

(2) $u = v \Rightarrow v_x' = 0$ $\Rightarrow l' = l_0$ since then $K' = K''$

Accelerated motion

Newtonian mechanics: $\vec{a}\,' = \vec{a}$

Relativity: $$dv_x = d\left(\frac{v_x' + V}{1 + \frac{v_x' V}{c^2}}\right) = \frac{dv_x'}{1 + \frac{v_x' V}{c^2}} -$$

$$- \frac{v_x' + V}{\left(1 + \frac{v_x' V}{c^2}\right)^2} \cdot \frac{dv_x' V}{c^2} =$$

$$= \frac{dv_x'}{1 + \frac{v_x' V}{c^2}} \left(1 - \frac{\frac{V}{c^2}(v_x' + V)}{1 + \frac{v_x' V}{c^2}}\right) = \frac{(1 - V^2/c^2)\, dv_x'}{\left(1 + \frac{v_x' V}{c^2}\right)^2}$$

$$\Rightarrow dv_x = \frac{dv_x'}{\gamma^2 \left(1 + \frac{v_x'}{c}\beta\right)^2}$$

$$a_x = \frac{dv_x}{dt} = \frac{\frac{dv_x'}{\gamma^2\left(1 + \frac{v_x'\beta}{c}\right)^2}}{\gamma\left(dt' + \frac{\beta}{c}dx'\right)} = \frac{\frac{dv_x'}{dt'}}{\gamma^3\left(1 + \frac{v_x'\beta}{c}\right)^3}$$

$$\Rightarrow a_x = \frac{a_x'}{\gamma^3\left(1 + \frac{v_x'\beta}{c}\right)^3}$$

Motion with constant acceleration is not possible since it will lead to v_x exceeding c! But we can imagine motion where acceleration is constant in the <u>instantaneous rest frame</u> of the particle

This is a frame K' where $\vec{v}' = 0$, although acceleration $\vec{a}' = \dot{\vec{v}}' \neq 0$. As particle moves, it goes through an infinite sequence of rest frames in each of which its $\vec{v}' = 0$.

Thus $a'_x = a_0$, $v'_x = 0$.

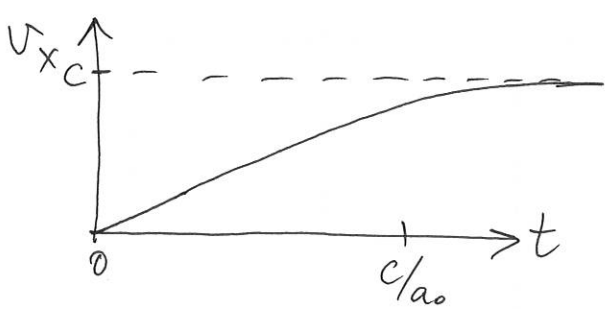
$$\Rightarrow \quad \frac{dv_x}{dt} = a_x = \cancel{\frac{a_0}{\left(1-\frac{v_x^2}{c^2}\right)^{3/2}}} \quad \frac{a_0}{\gamma^3} = a_0\left(1-\frac{v_x^2}{c^2}\right)^{3/2}$$

$$\Rightarrow \quad a_0 t = \int dv_x \left(1-\frac{v_x^2}{c^2}\right)^{-3/2} =$$

$$= \int_0^v \frac{dv_x}{\left(1-\frac{v_x^2}{c^2}\right)^{3/2}} = \frac{v_x}{\left(1-\frac{v_x^2}{c^2}\right)^{1/2}}$$

So $\dfrac{v_x^2}{1-\frac{v_x^2}{c^2}} = a_0^2 t^2$, $\quad v_x^2\left(1+\frac{a_0^2 t^2}{c^2}\right) = a_0^2 t^2$

$$v_x = \frac{a_0 t}{\sqrt{1+\frac{a_0^2 t^2}{c^2}}}$$



Thus at small $t \ll \frac{c}{a_0}$ $\quad v_x \approx a_0 t$ (non-relativistic motion)

but approaches $v_x \approx c$ for $t \gg \frac{c}{a_0}$